UNIT-1

Q1. What is a PN junction diode and how is it constructed?

A semiconductor diode is created by joining a P-type semiconductor material (having majority hole carriers) to an N-type semiconductor material (having majority electron carriers) within a single crystal structure. The P-region terminal is called the Anode and the N-region terminal is called the Cathode.

Q2. Explain the formation of the depletion region and barrier potential in an unbiased PN junction.

When a PN junction is formed without an applied bias, free electrons diffuse across the junction from the N-side to the P-side to recombine with holes. This leaves behind a layer of uncovered positive donor ions on the N-side and uncovered negative acceptor ions on the P-side, creating a carrier-depleted space-charge layer (depletion region) and establishing an internal barrier potential (V_0) that halts further diffusion.

Q3. State the cut-in (threshold) voltage values for Germanium (Ge) and Silicon (Si) diodes.

The cut-in or threshold voltage (V_K) below which the forward current is extremely small and above which current rises rapidly is approximately $0.3V$ **for Germanium (Ge)** and $0.7V$ **for Silicon (Si)**.

Q4. Write the Shockley diode current equation and define its variables.

The general current-voltage relation of a semiconductor diode is given by:

$$I_D = I_S\left(e^{\frac{V_D}{\eta V_T}} - 1\right)$$

where I_D is diode current, I_S is reverse saturation current, V_D is applied diode voltage, η is the ideality factor ($\eta = 1$ for Ge, $\eta = 2$ for Si at low currents), and $V_T = \frac{kT}{q}$ is the thermal voltage ($\approx 26mV$ at $300K$).

Q5. How does the depletion region width change under forward-bias and reverse-bias conditions?

Under forward bias (positive supply to P-type, negative to N-type), majority carriers are pushed toward the junction, reducing the barrier height and narrowing the depletion region. Under reverse bias, majority carriers are pulled away from the junction, increasing space charge and widening the depletion region.

Q6. What is transition capacitance (C_T) and in which biasing region is it dominant?

Transition capacitance (C_T) arises in the **reverse-bias region** where the non-conducting depletion layer acts as a dielectric insulator separating two layers of opposite immobile space charges, behaving like a parallel-plate capacitor ($C_T = \frac{\epsilon A}{W}$).

Q7. What is diffusion capacitance (C_D) and what formula defines it?

Diffusion capacitance (C_D) is the dominant capacitance in the **forward-bias region** caused by the storage of injected excess minority carriers in the quasi-neutral regions outside the depletion layer. It is defined as:

$$C_D = \frac{dQ}{dV} = g\tau_T = \frac{\tau_T I_D}{\eta V_T}$$

where τ_T is minority carrier lifetime and g is dynamic conductance.

Q8. How does applied reverse-bias voltage affect the transition capacitance of a diode?

Increasing reverse-bias voltage (V_R) widens the depletion layer width (W). Because capacitance is inversely proportional to depletion width ($C_T \propto \frac{1}{W}$), **increasing reverse bias decreases the transition capacitance**.

Q9. What is a Zener diode and how is it biased in voltage regulator circuits?

A Zener diode is a heavily doped silicon PN junction diode designed to operate safely in its reverse breakdown region without damage. In voltage regulator circuits, it is operated under **reverse-bias conditions** beyond its Zener potential (V_Z), maintaining a constant terminal voltage.

Q10. Differentiate between Zener breakdown and Avalanche breakdown.

Zener Breakdown: Occurs at low reverse voltages ($< 5V$) in heavily doped, narrow junctions where a strong electric field directly ruptures covalent bonds.

Avalanche Breakdown: Occurs at higher reverse voltages in lightly doped junctions where high-velocity minority carriers acquire sufficient kinetic energy to liberate bound electrons via impact collisions.

Q11. Define rectification and state the DC output voltage formula for a Half-Wave Rectifier.

Rectification is the process of converting an AC sinusoidal waveform of zero average value into a pulsating unidirectional waveform containing a DC level. For a half-wave rectifier with peak input voltage V_m, the average DC voltage is:

$$V_{dc} = 0.318V_m = \frac{V_m}{\pi}$$

Q12. What is the ripple factor (r) and what are its values for Half-Wave and Full-Wave rectifiers?

The ripple factor measures the effectiveness of a rectifier in converting AC to DC and is defined as the

ratio of the RMS value of the AC ripple component to the DC output voltage ($r = \frac{V_{r(rms)}}{V_{dc}}$).

- **Half-Wave Rectifier**: Ripple factor $r = 1.21$ (121%).
- **Full-Wave Rectifier**: Ripple factor $r = 0.48$ (48%).

Q13. Compare the Peak Inverse Voltage (PIV) ratings of a Center-Tapped Full-Wave Rectifier and a Bridge Rectifier.

- **Bridge Rectifier**: $PIV \geq V_m$ (where V_m is the peak secondary voltage).
- **Center-Tapped Rectifier**: $PIV \geq 2V_m$ (due to series addition of peak transformer secondary half-voltage and capacitor load voltage).

Q14. Explain the light emission mechanism in an LED and state why Silicon and Germanium cannot be used.

In a forward-biased LED, injected minority carriers recombine with majority carriers near the junction, releasing energy as visible or infrared photons. **Si and Ge cannot be used** because recombination energy is dissipated predominantly as heat within the lattice rather than light. Compound semiconductors like Gallium Arsenide (GaAs) are used instead.

Q15. Compare Light-Emitting Diodes (LEDs) and Liquid Crystal Displays (LCDs) in terms of light generation and power consumption.

- **LEDs**: Emit light directly when forward-biased, consuming power in the milliwatt (mW) range.
- **LCDs**: Do not generate their own light (they reflect or transmit light from an internal/external light source), consuming much lower power in the microwatt (\(\mu\text{W}\)) range.

Q16. What is a solar cell and how does it generate electrical power?

A solar cell is an un-biased PN junction device that converts solar photon light energy directly into electrical energy. Incident photons collide with valence atoms to generate electron-hole pairs near the junction, which are swept across by the internal electric field to produce a reverse photocurrent (I_{sc}).

Q17. Why are laser diodes often preferred over standard LEDs in fiber-optic communications?

Laser diodes produce coherent light that operates at higher data transmission rates, supports higher output power, and exhibits lower coupling and transmission losses into fiber cables compared to LEDs.

Q18. What is a Schottky barrier diode and state one major operational advantage over a standard PN junction diode.

A Schottky barrier diode is formed by a metal-semiconductor junction (e.g., gold or platinum on N-type silicon) where conduction occurs exclusively via majority carriers ("hot carriers"). It features a **lower forward threshold voltage ($\approx 0.2V$)** and **virtually zero reverse recovery time**, making it ideal for high-frequency switching and low-voltage power converters.

Q19. Describe the structure of a PIN diode and mention its primary application.

A PIN diode consists of a high-resistivity **Intrinsic (i) semiconductor layer** sandwiched between heavily doped P-type and N-type boundary layers. It is primarily used as a high-frequency RF switch, attenuator, or surface-mount high-power RF component.

Q20. What is a point-contact diode?

A point-contact diode is constructed by pressing a fine metallic wire point (cat whisker) against an N-type semiconductor wafer, creating a small-area junction that minimizes parasitic capacitance for ultra-high-frequency RF detection and mixing.

UNIT_2

Q1. Describe the structural differences and doping profile among the three terminals of a BJT.

- **Emitter (E)**: **Heavily doped** to inject majority carriers into the base.
- **Base (B)**: **Lightly doped** and **physically very thin** to allow carriers to diffuse across with minimal recombination.
- **Collector (C)**: **Moderately doped** and physically larger to dissipate thermal heat generated during device operation.

Q2. State the junction biasing conditions for a BJT operating in the Active, Cutoff, and Saturation regions.

- **Active Region**: Base–Emitter (EB) junction is **forward-biased**, Collector–Base (CB) junction is **reverse-biased**.
- **Cutoff Region**: Both EB and CB junctions are **reverse-biased** ($I_C \approx 0A$).
- **Saturation Region**: Both EB and CB junctions are **forward-biased** ($V_{CE,sat} \approx 0.2V$).

Q3. Derive the mathematical relationship between Common-Base current gain (α) and Common-Emitter current gain (β).

Using the terminal current equation $I_E = I_C + I_B$, divide both sides by I_C: \[\frac{I_E}{I_C} = 1 + \frac{I_B}{I_C} \implies \frac{1}{\alpha} = 1 + \frac{1}{\beta} = \frac{\beta + 1}{\beta}\]

Inverting both sides yields:

$$\alpha = \frac{\beta}{\beta+1} and \beta = \frac{\alpha}{1-\alpha}$$

Q4. What is the Early Effect (Base-Width Modulation) in a BJT, and how does it affect output characteristics?

As the reverse-bias voltage across the collector-base junction (V_{CB} or V_{CE}) increases, the depletion layer widens into the base region, reducing the effective base width. This decreases carrier recombination in the base, causing collector current (I_C) to increase slightly with increasing V_{CE} and introducing a finite small-signal output resistance ($r_o = \frac{V_A}{I_C}$).

Q5. Explain why a JFET is called a voltage-controlled unipolar device and why its gate current (I_G) is essentially zero.

A JFET is **unipolar** because conduction occurs strictly through majority carriers in a single channel (N-type or P-type). It is **voltage-controlled** because applied gate-to-source voltage (V_{GS}) creates an electric field that varies depletion width to regulate drain current (I_D). Because the gate-channel $P-N$ junction is kept reverse-biased during operation, gate current is virtually zero ($I_G \approx 0A$).

Q6. Define Pinch-off Voltage (V_P) and state Shockley's transfer equation for a JFET.

Pinch-off voltage (V_P) is the value of drain-to-source voltage (V_{DS}) at $V_{GS} = 0V$ (or gate voltage $V_{GS(off)}$) at which the reverse-biased depletion regions touch, causing drain current to saturate at I_{DSS}. In saturation, JFET transfer behavior follows **Shockley's equation**:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Q7. Derive the equation for maximum transconductance (g_{m0}) of a JFET in terms of I_{DSS} and V_P.

Differentiating Shockley's equation $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$ with respect to V_{GS}:

$$g_m = \frac{dI_D}{dV_{GS}} = \frac{2I_{DSS}}{|V_P|}\left(1 - \frac{V_{GS}}{V_P}\right)$$

Evaluating at $V_{GS} = 0V$ gives maximum transconductance:

$$g_{m0} = \frac{2I_{DSS}}{|V_P|}$$

Q8. How does a JFET function as a Voltage-Controlled Resistor (VCR) in its ohmic region?

At low drain-to-source voltages ($V_{DS} \ll |V_P|$), the channel has not reached pinch-off and behaves as a linear resistance (r_d) whose value is controlled by V_{GS}:

$$r_d = \frac{r_0}{\left(1-\frac{V_{GS}}{V_P}\right)^2}$$

where r_0 is the channel resistance at $V_{GS} = 0V$.

Q9. Differentiate between Depletion-type MOSFET (D-MOSFET) and Enhancement-type MOSFET (E-MOSFET).

- **D-MOSFET**: Constructed with a **physical built-in channel**. Operates in both **depletion mode** (negative V_{GS}) and **enhancement mode** (positive V_{GS}), conducting current at $V_{GS} = 0V$ ($I_D = I_{DSS}$).
- **E-MOSFET**: Constructed **without a built-in channel**. Operates strictly in **enhancement mode**, requiring V_{GS} to exceed a positive threshold voltage (V_T) to induce a channel ($I_D = 0A$ at $V_{GS} = 0V$).

Q10. Define Threshold Voltage (V_T) in an E-MOSFET and explain how an inversion layer is formed.

Threshold voltage (V_T) is the minimum gate-to-source voltage required to induce a conducting channel. When $V_{GS} > V_T$, positive electric potential at the gate repels majority holes from the SiO_2-substrate interface and attract minority electrons, forming an **n-type inversion layer** connecting source to drain.

Q11. What is Channel-Length Modulation in a MOSFET and how does it affect output resistance (r_o)?

In saturation ($V_{DS} > V_{GS} - V_T$), increasing V_{DS} causes the pinch-off point to shift toward the source, shortening effective channel length from L to $L - \Delta L$. This increases drain current slightly according to $I_D = K_N(V_{GS} - V_T)^2(1 + \lambda V_{DS})$, resulting in a finite small-signal output resistance $r_o = \frac{1}{\lambda I_D} = \frac{V_E}{I_D}$.

Q12. What are VMOS and UMOS (Trench) MOSFETs, and why do they offer higher power capabilities than planar MOSFETs?

VMOS and UMOS are power MOSFETs constructed with V-shaped or U-shaped (trench) vertical gate structures. Current flows **vertically** through the substrate rather than horizontally, shortening channel length, lowering ON-resistance ($R_{DS(on)}$), and providing wider conduction areas that support higher drain currents ($> 1A$) and power handling ($> 10W$).

Q13. What is a Metal-Semiconductor Field-Effect Transistor (MESFET) and how does its gate barrier differ from a MOSFET?

A MESFET is a field-effect transistor that replaces the SiO_2 oxide insulator of a MOSFET with a direct Schottky metal-semiconductor contact (e.g., tungsten or gold deposited on an n-type GaAs channel) to form the gate junction.

Q14. State two operational advantages of GaAs MESFETs over Silicon MOSFETs in high-frequency RF applications.

1. **Higher Electron Mobility**: Gallium Arsenide (GaAs) has significantly higher electron mobility than silicon, enabling faster switching speeds and higher cut-off frequencies.
2. **Lower Stray Capacitance**: Eliminating the oxide layer reduces gate-to-channel parasitic capacitance, reducing high-frequency signal shunting.

Q15. What is a FinFET? Describe its 3D gate construction.

A FinFET is a 3D multi-gate transistor structure where the conducting channel is formed by a thin vertical semiconductor "fin" (silicon ridge). The gate electrode wraps around **three sides of the fin channel**, providing multi-directional electrostatic control over carrier flow.

Q16. Why are FinFETs preferred over planar MOSFETs in nanoscale (sub-10nm) semiconductor nodes?

Wrapping the gate around three sides of the vertical fin gives superior electrostatic control, suppressing unwanted **short-channel effects (SCE)**, reducing subthreshold off-state leakage current, and enabling lower operating supply voltages at scaled technology nodes.

Q17. Describe the construction of a Unijunction Transistor (UJT) and define its Intrinsic Standoff Ratio (η).

A UJT consists of a lightly doped n-type silicon bar with two ohmic base contacts (B_1 and B_2) and a single p-n junction formed by alloying an aluminum rod (Emitter, E). Its **Intrinsic Standoff Ratio (η)** is the internal resistive voltage-divider ratio:

$$\eta = \frac{R_{B1}}{R_{B1}+R_{B2}} = \frac{R_{B1}}{R_{BB}} \; (0.51 \leq \eta \leq 0.82)$$

Q18. Write the expression for the Emitter Peak Firing Voltage (V_P) of a UJT and explain its negative-resistance region.

The firing voltage is $V_P = \eta V_{BB} + V_D$ (where $V_D \approx 0.7V$). When emitter voltage V_E exceeds V_P, hole injection into the B_1 region dramatically lowers resistance R_{B1}, causing emitter voltage to drop as emitter current increases, creating a **stable negative-resistance region** ideal for relaxation oscillators.

Q19. Describe the four-layer PNPN structure of a Silicon-Controlled Rectifier (SCR) and explain its operation using the two-transistor model.

An SCR is a 4-layer (PNPN), 3-terminal (Anode, Cathode, Gate) bistable semiconductor switch. It can be modeled as interconnected PNP (Q_1) and NPN (Q_2) transistors in a regenerative feedback loop; applying a positive gate current pulse triggers Q_2 into conduction, driving Q_1 on and locking the SCR into a low-impedance ON state.

Q20. Differentiate between Anode Current Interruption and Forced Commutation methods for turning off an SCR.

- **Anode Current Interruption**: Opens the series anode circuit (or shunts current around the SCR) to reduce anode current below the holding current ($I_A < I_H$).
- **Forced Commutation**: Applies a reverse voltage (via a switching capacitor or transistor circuit) across the conducting SCR to force current in the reverse direction, turning it OFF.

UNIT_3

Q1. What is small-signal linearization in a BJT, and what is the mathematical condition for small-signal operation?

The BJT exponential current-voltage relation $i_C = I_S e^{v_{BE}/V_T}$ is non-linear. Expanding via Taylor series around the DC quiescent point yields $i_C \approx I_{CQ}(1 + \frac{v_{be}}{V_T})$. For linear operation with negligible harmonic distortion, the AC input voltage must satisfy the **small-signal condition**:

$$v_{be} \ll V_T (V_T \approx 26mVatT = 300K)$$

Q2. Define small-signal transconductance (g_m) and input base resistance (r_π) of a BJT in terms of DC bias parameters.

- **Transconductance (g_m)**: \(g_m = \left.\frac{\partial i_C}{\partial v_{BE}}\right|_{I_{CQ}} = \frac{I_{CQ}}{V_T} \quad [\text{A/V or mhos}]\)
- **Input Base Resistance (r_π)**: \(r_\pi = \left.\frac{\partial v_{BE}}{\partial i_B}\right|_{I_{BQ}} = \frac{\beta V_T}{I_{CQ}} = \frac{\beta}{g_m} = \beta r_e\)

Q3. Derive the voltage gain (A_v) of a standard Common-Emitter (CE) amplifier with a bypassed emitter resistor using small-signal parameters.

In a CE amplifier with C_E shorted at midband:

Output voltage: $V_o = -g_m v_\pi (R_C \parallel r_o) \approx -g_m v_\pi R_C$

Input voltage: $V_i = v_\pi$

Voltage gain:

$$A_v = \frac{V_o}{V_i} = -g_m (R_C \parallel r_o) \approx -g_m R_C = -\frac{R_C}{r_e}$$

(The negative sign indicates a 180° ***phase inversion****).*

Q4. How does an unbypassed emitter resistor (R_E) affect the voltage gain and input resistance of a CE amplifier?

An unbypassed R_E introduces negative feedback:

- **Input Resistance**: $Z_b = r_\pi + (1 + \beta)R_E \approx r_\pi + \beta R_E$ (significantly increased).
- **Voltage Gain**: $A_v = \frac{-\beta R_C}{r_\pi + (1+\beta)R_E} \approx -\frac{R_C}{R_E}$ (reduced, but stabilized against β and V_T temperature variations).

Q5. State the voltage gain (A_v), input resistance (R_{in}), and phase shift for a Common-Base (CB) amplifier configuration.

- **Voltage Gain**: $A_v = +g_m R_C = +\frac{R_C}{r_e}$ (High, non-inverting).
- **Input Resistance**: $R_{in} = r_e = \frac{\alpha}{g_m} \approx \frac{1}{g_m}$ (Very Low, $\approx 20, \Omega - 100, \Omega$).
- **Phase Shift**: 0° (Non-inverting).

Q6. Why is the Common-Collector (CC) amplifier called an Emitter Follower, and what are its typical gain and impedance characteristics?

It is called an Emitter Follower because the output at the emitter follows the input signal applied to the base in amplitude and phase:

- **Voltage Gain**: $A_v = \frac{(1+\beta)R_L}{r_\pi + (1+\beta)R_L} \approx 1$ ($A_v < 1$, unity gain buffer).
- **Input Resistance**: $R_{in} \approx r_\pi + \beta R_L$ (Very High).
- **Output Resistance**: $R_{out} \approx r_e + \frac{R_{sig}}{\beta+1} \approx \frac{1}{g_m}$ (Very Low).

Q7. Compare the performance parameters of CE, CB, and CC amplifier configurations.

- **CE**: Moderate R_{in} ($\sim k\Omega$), Moderate R_{out} ($\sim R_C$), High A_v (inverting 180°), High A_i.
- **CB**: Low R_{in} ($\sim r_e$), High R_{out} ($\sim R_C$), High A_v (0°), Unity A_i ($\alpha \approx 1$).
- **CC**: High R_{in} ($\sim \beta R_L$), Low R_{out} ($\sim 1/g_m$), Unity A_v ($A_v \approx 1,0^\circ$), High A_i.

Q8. What factors dictate the low-frequency response of a BJT amplifier, and how is the lower cutoff frequency (f_{Ls}) due to input coupling capacitor C_s derived?

The low-frequency response is governed by external coupling capacitors (C_s, C_C) and bypass capacitor (C_E). The cutoff frequency due to C_s is determined by the series RC time constant of the input loop:

$$f_{Ls} = \frac{1}{2\pi(R_s+R_i)C_s}$$

where $R_i = R_1 \parallel R_2 \parallel r_\pi$.

Q9. Derive the lower cutoff frequency (f_{LE}) contributed by the emitter bypass capacitor (C_E).

Looking back into the emitter terminal, the equivalent resistance R_{eq} seen by C_E is:

$$R_{eq} = R_E \parallel \left[r_e + \frac{R_s \parallel R_1 \parallel R_2}{\beta}\right]$$

$$f_{LE} = \frac{1}{2\pi R_{eq} C_E}$$

(Since R_{eq} is small, f_{LE} is typically the highest lower-frequency cutoff pole and dominates f_L).

Q10. Define the BJT beta cutoff frequency (f_β) and transition frequency (f_T), and state their mathematical relationship.

Beta Cutoff Frequency (f_β): The frequency at which the short-circuit AC current gain h_{fe} drops by $3dB$ (0.707) from its midband value:

$$f_\beta = \frac{1}{2\pi h_{fe,mid} r_e (C_{be}+C_{bc})}$$

Transition Frequency (f_T): The unity-gain frequency where $|h_{fe}| = 1$:

$$f_T = h_{fe,mid} \cdot f_\beta = \frac{1}{2\pi r_e (C_{be}+C_{bc})} = \frac{g_m}{2\pi(C_\pi + C_\mu)}$$

Q11. Derive the small-signal transconductance (g_m) of an Enhancement MOSFET (E-MOSFET) operating in the saturation region.

The saturation drain current is $i_D = K_N(v_{GS} - V_T)^2$. Differentiating i_D with respect to v_{GS} at constant v_{DS}: \[g_m = \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{V_{DS}} = \frac{d}{dv_{GS}} \left[K_N (v_{GS} - V_T)^2 \right] = 2 K_N (V_{GSQ} - V_T)\] Substituting $V_{GSQ} - V_T = \sqrt{\frac{I_{DQ}}{K_N}}$ yields:

$$g_m = 2\sqrt{K_N I_{DQ}}$$

Q12. What is channel-length modulation in a MOSFET, and how is the small-signal output resistance (r_d or r_o) derived?

Beyond saturation ($v_{DS} > v_{GS} - V_T$), increasing v_{DS} shortens the effective channel length L, adding a slope parameter λ ($i_D = K_N(v_{GS} - V_T)^2(1 + \lambda v_{DS})$). Differentiating yields: \[r_d = r_o = \left(\left.\frac{\partial i_D}{\partial v_{DS}}\right|_{V_{GSQ}} \right)^{-1} = \frac{1}{\lambda K_N (V_{GSQ} - V_T)^2} \approx \frac{1}{\lambda I_{DQ}} = \frac{V_E}{I_{DQ}}\]

Q13. Derive the midband voltage gain (A_v) and input resistance (R_{in}) of a Common-Source (CS) MOSFET amplifier with voltage-divider bias.

Input Resistance: Gate DC current is zero ($I_G = 0A$), so $R_i = \infty$. Including bias resistors R_1, R_2:

$$R_{in} = R_1 \parallel R_2 = R_G$$

Voltage Gain: $V_o = -g_m v_{gs}(r_d \parallel R_D)$ and $V_i = v_{gs}$:

$$A_v = \frac{V_o}{V_i} = -g_m(r_d \parallel R_D) \approx -g_m R_D (for r_d \gg R_D)$$

Q14. Derive the output resistance (Z_o) of a Common-Source (CS) MOSFET amplifier.

Set the independent signal source $V_i = 0V$. This makes $v_{gs} = 0V$, causing the controlled current source $g_m v_{gs} = 0A$ (open circuit). Looking into the output drain node:

$$Z_o = r_d \parallel R_D \approx R_D (when r_d \geq 10R_D)$$

Q15. Why is a Common-Drain (CD) MOSFET configuration called a Source Follower, and what is its voltage gain expression?

It is called a **Source Follower** because the output voltage at the source terminal follows the gate input voltage in phase (0° phase shift) with near-unity gain. Applying KVL ($V_i = v_{gs} + V_o$) and KCL ($V_o = g_m v_{gs}(r_d \parallel R_S)$):

$$A_v = \frac{V_o}{V_i} = \frac{g_m(r_d \| R_S)}{1+g_m(r_d \| R_S)} \approx \frac{g_m R_S}{1+g_m R_S} < 1$$

Q16. Derive the small-signal output resistance (Z_o) of a Source Follower (Common Drain) amplifier.

Set input $V_i = 0V$ (grounding gate node) and apply a test voltage V_x at the source node ($v_{gs} = -V_x$). Applying KCL at the source node: \[I_x + g_m v_{gs} = \frac{V_x}{r_d} + \frac{V_x}{R_S} \implies I_x = V_x \left(g_m + \frac{1}{r_d} + \frac{1}{R_S} \right)\]

$$Z_o = \frac{V_x}{I_x} = \frac{1}{g_m} \parallel r_d \parallel R_S \approx \frac{1}{g_m} \parallel R_S \approx \frac{1}{g_m}$$

Q17. Explain the Miller Effect and state the mathematical expressions for Miller input capacitance (C_{Mi}) and output capacitance (C_{Mo}).

The **Miller Effect** describes the multiplication of feedback capacitance C_f bridging the input and output terminals of an inverting amplifier (gain $-A_v$):

- **Input Miller Capacitance**: $C_{Mi} = (1 - A_v)C_f = (1 + |A_v|)C_{gd}$
- **Output Miller Capacitance**: $C_{Mo} = \left(1 - \frac{1}{A_v}\right)C_f \approx C_{gd} (for |A_v| \gg 1)$

Q18. Identify the internal parasitic capacitances that limit the high-frequency response of a MOSFET.

1. **Gate-to-Source Capacitance (C_{gs})**: Interelectrode gate-channel depletion/accumulation capacitance.
2. **Gate-to-Drain Capacitance (C_{gd})**: Overlap capacitance between gate and drain, which undergoes Miller multiplication.
3. **Drain-to-Source Capacitance (C_{ds})**: Junction capacitance between drain-body substrate.

Q19. How is the upper $3dB$ cutoff frequency (f_{Hi}) of a Common-Source MOSFET amplifier determined using Miller's theorem?

Summing total input capacitance $C_i = C_{Wi} + C_{gs} + C_{Mi} = C_{Wi} + C_{gs} + (1 + |A_v|)C_{gd}$ and determining Thevenin input resistance $R_{Thi} = R_{sig} \parallel R_G$:

$$f_{Hi} = \frac{1}{2\pi R_{Thi} C_i} = \frac{1}{2\pi(R_{sig} \| R_G)\left[C_{Wi} + C_{gs} + (1+|A_v|)C_{gd}\right]}$$

Q20. Compare the high-frequency performance of Common-Source (CS) and Common-Gate (CG) / Common-Base (CB) amplifiers regarding the Miller effect.

- **Common-Source (CS)**: Inverting gain ($-A_v$) causes severe Miller multiplication of C_{gd}, creating a large input capacitance C_{Mi} that significantly

lowers f_{Hi} and limits high-frequency bandwidth.

- **Common-Gate (CG) / Common-Base (CB)**: Non-inverting gain ($+A_v$) with grounded gate/base eliminates Miller multiplication of feedback capacitance, offering vastly superior high-frequency response and wider bandwidth.

UNIT-4

Q1. What is a differential amplifier, and how are difference-mode and common-mode input voltages defined?

A differential amplifier is a circuit designed to amplify the voltage difference between two input terminals while suppressing unwanted signals common to both.

- **Difference-mode input voltage (V_d)**: $V_d = V_1 - V_2$
- **Common-mode input voltage (V_c)**: $V_c = \frac{V_1+V_2}{2}$

Q2. Write the expression for the net output voltage of a practical differential amplifier considering both differential and common-mode gains.

The total output voltage is given by:

$$V_o = A_d V_d + A_c V_c$$

where A_d is the differential-mode voltage gain, A_c is the common-mode voltage gain, V_d is the differential input voltage, and V_c is the common-mode input voltage.

Q3. Define Common-Mode Rejection Ratio (CMRR) and state its mathematical expression in linear and decibel (dB) forms.

CMRR is a figure of merit quantifying the ability of a differential amplifier to amplify difference signals while rejecting common-mode noise.

- **Linear ratio**: $CMRR = \left|\frac{A_d}{A_c}\right|$
- **Decibel scale**: \(\text{CMRR}{\text{dB}} = 20 \log{10} \left| \frac{A_d}{A_c} \right|\) *(For an ideal differential amplifier, $A_c = 0$, giving $CMRR = \infty$).*

Q4. Why does the common-emitter node act as a virtual AC ground during differential-mode operation?

In differential-mode operation, equal and opposite input signals ($V_1 = +\frac{V_s}{2}, V_2 = -\frac{V_s}{2}$) drive matched transistors Q_1 and Q_2. The resulting AC emitter currents are equal in magnitude but 180° out of phase ($i_{e1} = -i_{e2}$), making their sum at the shared emitter node zero ($v_e = 0V$), which functions as a **virtual AC ground**.

Q5. How does replacing the common-emitter resistor (R_E) with a constant-current source improve CMRR?

Common-mode voltage gain is inversely proportional to emitter resistance ($A_c \approx \frac{-R_C}{2R_E}$). Replacing passive R_E with an active constant-current source provides an extremely high AC internal resistance ($R_E = r_o \to \infty$) while dropping minimal DC voltage, driving $A_c \to 0$ and significantly increasing CMRR.

Q6. What is a single tuned amplifier, and what equation determines its resonant frequency (f_o)?

A single tuned amplifier is an RF amplifier stage that uses a parallel resonant LC tank circuit as its collector/drain load to select and amplify signals within a specific narrow frequency band. Its resonant frequency is given by:

$$f_o = \frac{1}{2\pi\sqrt{LC}}$$

Q7. Define the Quality Factor (Q) of a tuned LC tank circuit and explain its effect on amplifier selectivity.

The Quality Factor (Q) is the ratio of reactive power/reactance to internal series resistance at resonance ($Q = \frac{\omega_o L}{R} = \frac{1}{\omega_o CR}$). A higher Q increases the tank's resonant impedance and narrows the bandwidth ($BW = \frac{f_o}{Q}$), providing sharper frequency selectivity.

Q8. State the expressions for the voltage gain at resonance (A_{v0}) and bandwidth (BW) of a single tuned amplifier.

At resonance (f_o), the parallel LC tank reaches its maximum equivalent resistive impedance $R_p = Q^2 R = \frac{L}{CR}$.

- **Resonant Voltage Gain**: $A_{v0} = -g_m R_p$
- **Bandwidth**: $BW = f_H - f_L = \frac{f_o}{Q}$

Q9. How does the voltage gain of a single tuned amplifier vary as operating frequency moves away from resonance (f_o)?

At resonance (f_o), tank impedance is purely resistive and at its maximum, yielding peak voltage gain. At frequencies above or below f_o, capacitive or inductive reactance reduces the overall parallel tank impedance, causing voltage gain to drop off symmetrically on both sides of f_o.

Q10. What is the effect of cascading n identical single tuned stages on the overall amplifier bandwidth?

Cascading n identical single tuned stages shrinks the overall bandwidth due to cumulative gain roll-off at the corner frequencies. The overall bandwidth BW_n narrows according to:

$$BW_n = BW_1\sqrt{2^{1/n} - 1}$$

(For $n = 2$, the overall bandwidth drops to 64% of a single-stage bandwidth).

Q11. What is neutralization in high-frequency tuned amplifiers, and why is it necessary?

Neutralization is the process of feeding back an auxiliary AC signal equal in magnitude but 180° opposite in phase to cancel out internal parasitic feedback through the collector-base/gate-drain junction capacitance (C_{bc} or C_{gd}). It prevents self-oscillations and high-frequency instability.

Q12. How does the Miller effect contribute to instability in high-frequency RF amplifiers?

In an inverting amplifier, feedback capacitance C_{bc} (or C_{gd}) is multiplied at the input node by the voltage gain ($C_{Mi} = (1 + |A_v|)C_{bc}$). At high radio frequencies, phase shifts in this feedback branch convert negative feedback into positive feedback, causing the amplifier to oscillate.

Q13. Describe the Hazeltine neutralization method.

In Hazeltine neutralization, a center-tapped tuned secondary transformer or collector inductor provides a voltage that is 180° out of phase with the collector voltage. This phase-inverted voltage is fed back to the base through a neutralizing capacitor ($C_N = C_{bc}$), canceling out internal parasitic feedback current.

Q14. Describe the Rice neutralization method.

In Rice neutralization, the center-tapped inductor is placed across the **input (base/gate) circuit** instead of the output. The phase-inverted voltage developed across the tapped input coil is fed back to the collector/drain terminal through a neutralizing capacitor (C_N), neutralizing capacitive feedback.

Q15. How does using a Cascode configuration eliminate the need for neutralization?

A Cascode configuration cascades a Common-Emitter (or Common-Source) stage into a Common-Base (or Common-Gate) stage. Because the CB/CG stage presents a very low input impedance ($r_e \approx 1/g_m$), the first stage's voltage gain is held near unity ($|A_{v1}| \approx 1$), eliminating Miller capacitance multiplication ($C_{Mi} \approx 2C_{bc}$) and stabilizing the circuit without neutralizing networks.

Q16. Classify power amplifiers based on their operating conduction angle.

- Class A: Conducts for the full 360° (100%) of the input cycle.
- Class AB: Conducts between 180° and 360° of the input cycle.
- Class B: Conducts for exactly 180° (50%) of the input cycle.
- Class C: Conducts for less than 180° of the input cycle.
- Class D: Operates using pulse/digital switching signals.

Q17. Compare the maximum theoretical power efficiency (η) of Class A, Class B, Class AB, and Class D power amplifiers.

- **Class A**: Maximum $\eta = 25\%$ (series-fed) and 50% (transformer-coupled).

- **Class B**: Maximum $\eta = 78.5\%$ ($\frac{\pi}{4}$).
- **Class AB**: Efficiency ranges between $25\%/50\%$ **and** 78.5%.
- **Class D**: High efficiency, typically **over** 90%.

Q18. Explain the working principle of a Class B Push-Pull Amplifier and why two complementary transistors are required.

A single Class B transistor conducts for only 180°, creating severe waveform distortion. A push-pull amplifier uses two complementary transistors (NPN/PNP or nMOS/pMOS); one transistor "pushes" current during the positive half-cycle and the second "pulls" current during the negative half-cycle, combining to reconstruct the full 360° output waveform.

Q19. What is crossover distortion in Class B amplifiers, and how does Class AB operation eliminate it?

Crossover distortion occurs near zero-crossing because both transistors remain OFF until the input voltage exceeds the base-emitter threshold ($V_{BE} \approx 0.7V$). Class AB operation eliminates this by applying a slight DC forward bias to keep both transistors conducting slightly at zero input, ensuring smooth transitions between half-cycles.

Q20. Why are Class C power amplifiers restricted to tuned RF communications rather than audio amplification?

Class C amplifiers conduct in brief current pulses for less than 180°, generating extreme harmonic distortion unacceptable for audio. In RF communications, an LC resonant tank circuit connected to the load filters out harmonics and stores/releases energy to reconstruct a smooth sinusoidal waveform at the fundamental resonant frequency.

UNIT-5

Q1. What is negative feedback in an amplifier, and how does it affect closed-loop voltage gain?

Negative feedback occurs when a portion of the output signal is fed back to the input 180° out of phase with the applied signal. It reduces the net input signal, lowering overall voltage gain to:

$$A_f = \frac{A}{1+\beta A}$$

(where A is open-loop gain and β is feedback factor). In exchange, it stabilizes gain against parameter variations.

Q2. Differentiate between Voltage Feedback and Current Feedback networks.

- **Voltage Feedback**: Samples the output voltage by connecting the feedback network in **parallel (shunt)** with the load.
- **Current Feedback**: Samples the output current by connecting the feedback network in **series** with the load.

Q3. Differentiate between Series Mixing and Shunt Mixing at the input of a feedback amplifier.

- **Series Mixing**: The feedback signal is connected in series with the input signal source, adding/subtracting in **voltage** ($V_{in} = V_s - V_f$).
- **Shunt Mixing**: The feedback signal is connected in parallel with the input signal source, adding/subtracting in **current** ($I_{in} = I_s - I_f$).

Q4. How does Voltage-Series negative feedback affect the input and output impedances of an amplifier?

- **Input Impedance**: Increases by factor $(1 + \beta A)$: $Z_{if} = Z_i(1 + \beta A)$
- **Output Impedance**: Decreases by factor $(1 + \beta A)$: $Z_{of} = \frac{Z_o}{1+\beta A}$

Q5. How does Current-Series negative feedback affect the input and output impedances of an amplifier?

Input Impedance: Increases by factor $(1 + \beta A)$: $Z_{if} = Z_i(1 + \beta A)$

Output Impedance: Increases by factor $(1 + \beta A)$: $Z_{of} = Z_o(1 + \beta A)$

Q6. How does Voltage-Shunt negative feedback affect the input and output impedances of an amplifier?

Input Impedance: Decreases by factor $(1 + \beta A)$: $Z_{if} = \frac{Z_i}{1+\beta A}$

Output Impedance: Decreases by factor $(1 + \beta A)$: $Z_{of} = \frac{Z_o}{1+\beta A}$

Q7. What is the effect of negative feedback on amplifier bandwidth and Gain-Bandwidth Product (GBW)?

Negative feedback increases bandwidth by the factor $(1 + \beta A)$:

$$BW_f = BW(1 + \beta A)$$

Because lower cutoff frequency drops ($f_{Lf} = \frac{f_L}{1+\beta A}$) and upper cutoff frequency rises ($f_{Hf} = f_H(1 + \beta A)$), the **Gain-Bandwidth Product remains constant**.

Q8. What is positive (regenerative) feedback and how does it differ from negative feedback?

In positive feedback, the feedback signal is fed back **in phase** (0° or 360°) with the input signal. The feedback signal reinforces the input, increasing net loop gain ($A_f = \frac{A}{1-\beta A}$). When $1 - \beta A \to 0$, the circuit drives into self-sustained oscillations without an external AC input.

Q9. State the Barkhausen Criteria for sustained oscillations.

For an amplifier circuit to sustain continuous oscillations at a specific frequency f_o:

Magnitude Condition: The magnitude of the loop gain must equal unity:
$|\beta A| = 1$

Phase Condition: Total phase shift around the closed feedback loop must be an integer multiple of 360° (or 0°):

$\angle \beta A = 0^\circ or 360^\circ$

Q10. Why must loop gain $|\beta A|$ initially be set slightly greater than 1 during oscillator startup?

During startup, $|\beta A| > 1$ is required so that small internal thermal noise voltages can build up exponentially into oscillations. Once the desired output amplitude is reached, non-linear device saturation automatically lowers the loop gain to stable equilibrium ($|\beta A| = 1$).

Q11. Explain the working principle of an RC Phase-Shift Oscillator and state its frequency equation.

An RC Phase-Shift Oscillator uses an inverting amplifier (which provides 180° phase shift) paired with a 3-stage RC high-pass ladder feedback network. At the resonant frequency f_o, each RC section produces a phase shift such that the total RC ladder provides an additional 180° phase shift. Its frequency is:

$$f_o = \frac{1}{2\pi RC\sqrt{6}}$$

Q12. What is the minimum amplifier gain requirement for an RC Phase-Shift Oscillator, and why?

At the resonant frequency f_o, the attenuation through the 3-stage RC feedback network is $\beta = \frac{1}{29}$. To satisfy the Barkhausen magnitude condition $|\beta A| \geq 1$:

$|A| \geq 29$

*(The amplifier must provide an open-loop gain of at least **29** to overcome feedback network losses)*.

Q13. State the frequency equation and zero-phase-shift condition for a Wien Bridge Oscillator.

The reactive lead-lag arm of a Wien Bridge Oscillator consists of a series RC branch (R_1, C_1) and parallel RC branch (R_2, C_2). At resonance, it produces **zero phase shift** ($\angle \beta = 0^\circ$). For $R_1 = R_2 = R$ and $C_1 = C_2 = C$, the frequency is:

$$f_o = \frac{1}{2\pi RC}$$

Q14. What is the minimum gain requirement for the non-inverting amplifier in a Wien Bridge Oscillator?

At resonant frequency f_o, the lead-lag feedback network attenuation is $\beta = \frac{1}{3}$. To satisfy $|\beta A| \geq 1$, the non-inverting op-amp closed-loop gain must be: \[A_v = 1 + \frac{R_3}{R_4} \ge 3 \implies \frac{R_3}{R_4} \ge 2\]

Q15. Describe the tank circuit construction of a Hartley Oscillator and state its frequency formula.

A Hartley Oscillator uses a center-tapped inductive voltage divider consisting of two inductors (L_1, L_2) with mutual inductance M, connected in parallel with a single tuning capacitor C. Its resonant frequency is:

$$f_o = \frac{1}{2\pi\sqrt{L_{eq}C}} where L_{eq} = L_1 + L_2 + 2M$$

Q16. State the minimum gain requirement for a Hartley Oscillator.

The feedback factor is determined by the ratio of the split inductances ($\beta \approx \frac{L_1}{L_2}$). To satisfy Barkhausen criteria, the amplifier voltage gain must satisfy:

$$A_v \geq \frac{L_1+M}{L_2+M} \approx \frac{L_1}{L_2}$$

Q17. Describe the tank circuit construction of a Colpitts Oscillator and state its frequency formula.

A Colpitts Oscillator uses a capacitive voltage divider consisting of two split series capacitors (C_1, C_2) connected in parallel with a single tuning inductor L. Its resonant frequency is:

$$f_o = \frac{1}{2\pi\sqrt{LC_{eq}}} where C_{eq} = \frac{C_1C_2}{C_1+C_2}$$

Q18. State the minimum gain requirement for a Colpitts Oscillator.

The feedback fraction is set by the ratio of the capacitive divider ($\beta = \frac{C_2}{C_1}$). To satisfy the Barkhausen magnitude condition, the amplifier voltage gain must satisfy:

$$A_v \geq \frac{C_1}{C_2} (or h_{fe} \geq \frac{C_1}{C_2})$$

Q19. Describe the Piezoelectric Effect and draw the electrical equivalent circuit of a Quartz Crystal.

The **Piezoelectric Effect** occurs when mechanical stress applied across a crystal lattice induces an electric potential, and conversely, an applied AC voltage causes mechanical vibrations. Its electrical equivalent circuit consists of a **series** $R - L - C$ **branch** (representing mechanical mass, compliance, and friction) in parallel with a **mounting capacitance** C_M:

None

Q20. Define the Series (f_s) and Parallel (f_p) resonant frequencies of a Crystal Oscillator.

- **Series Resonant Frequency (f_s)**: Occurs when series branch reactances cancel ($X_L = X_C$), yielding minimum crystal impedance ($Z = R$):

$$f_s = \frac{1}{2\pi\sqrt{LC}}$$

- **Parallel (Antiresonant) Frequency (f_p)**: Occurs at a slightly higher frequency when the series RLC leg acts inductively and resonates with mounting capacitance C_M, producing peak crystal impedance:

$$f_p = \frac{1}{2\pi\sqrt{LC_{eq}}} where C_{eq} = \frac{CC_M}{C+C_M}$$